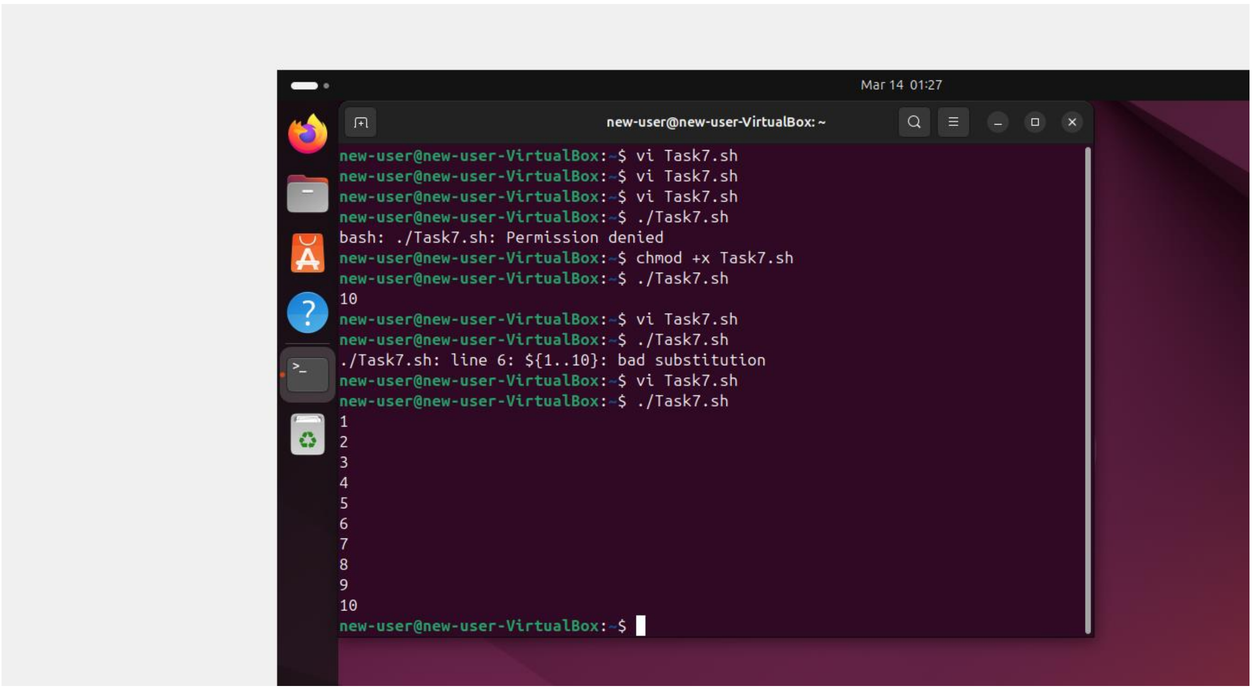
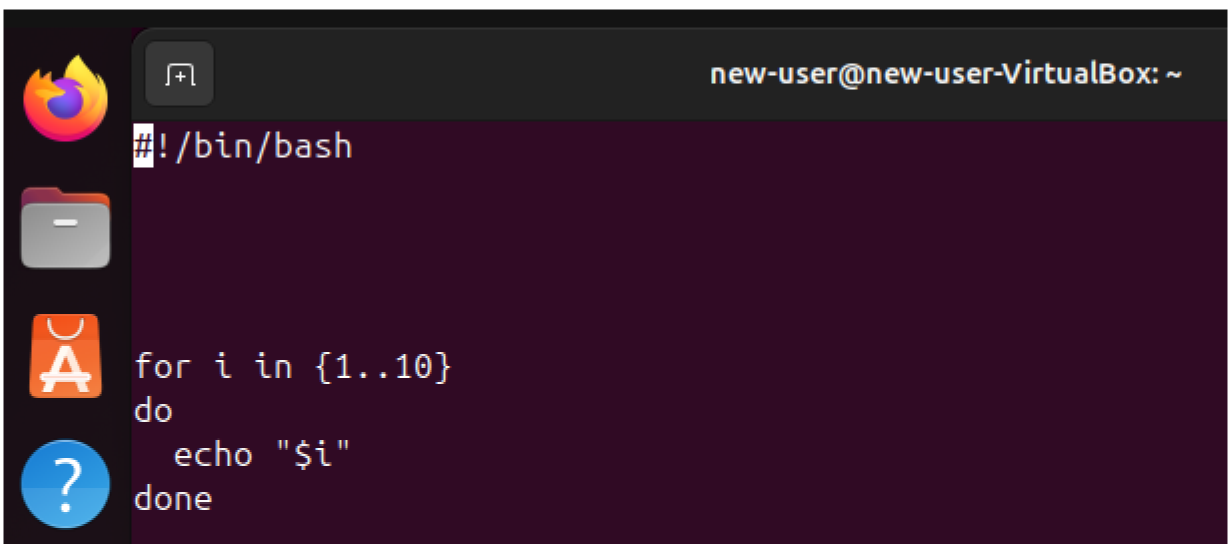
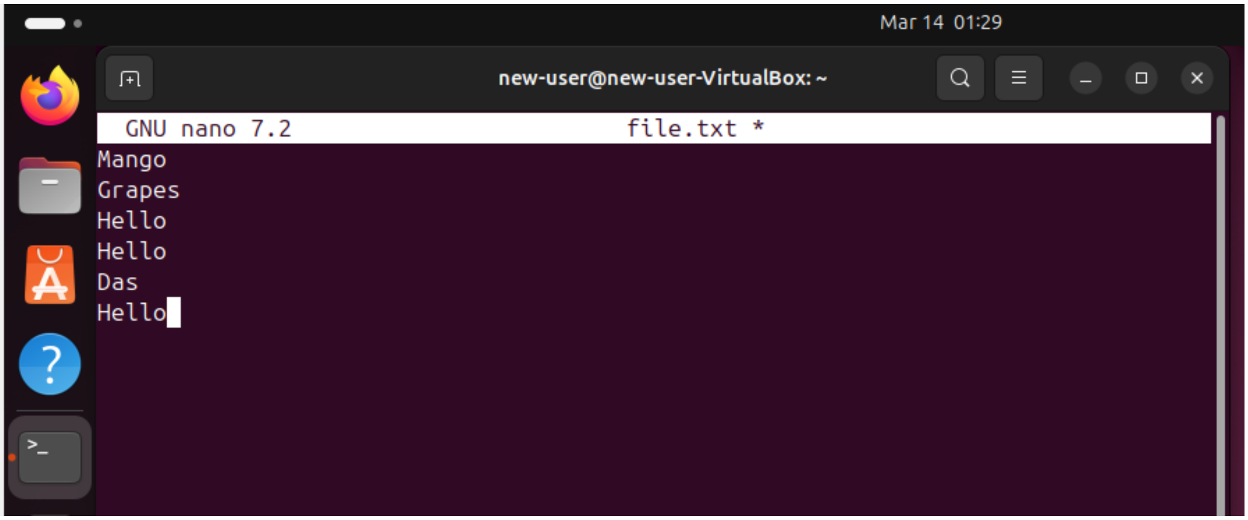
OS LAB 3 B TASKS

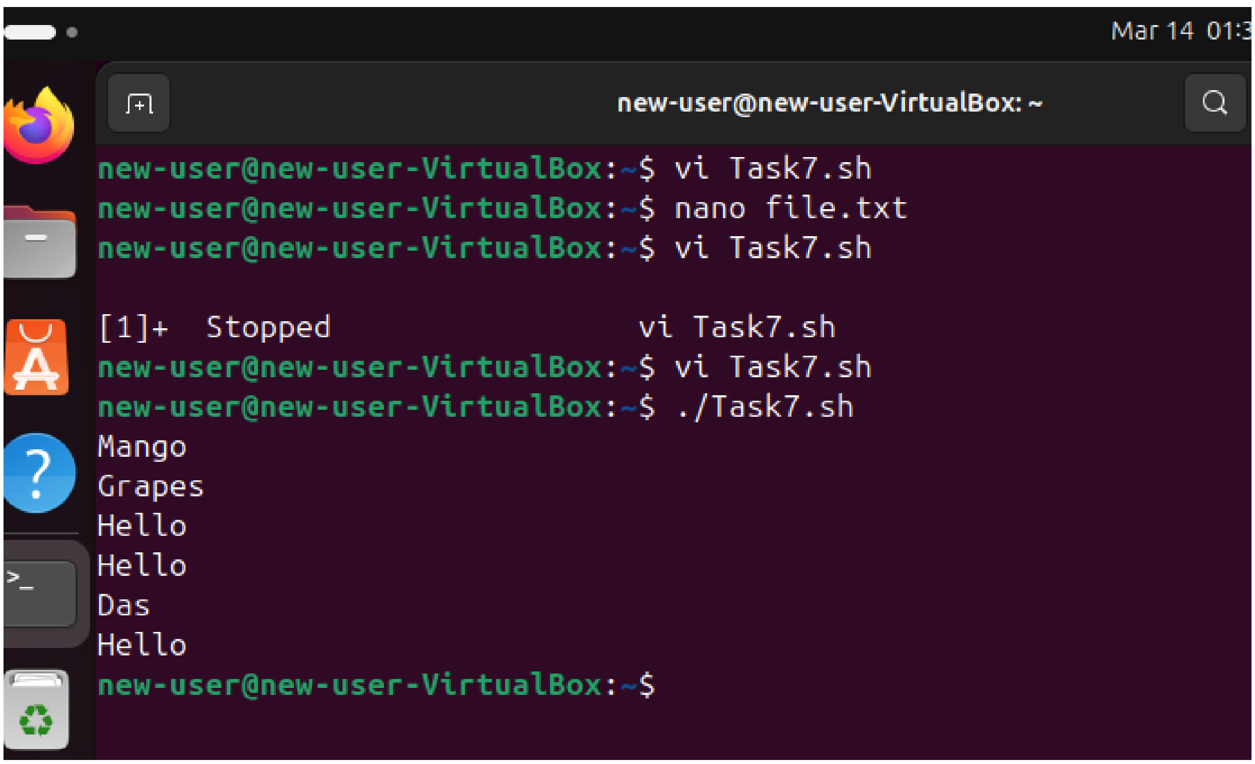
BAZIL UDDIN KHAN 24K-0559 BSCS4H

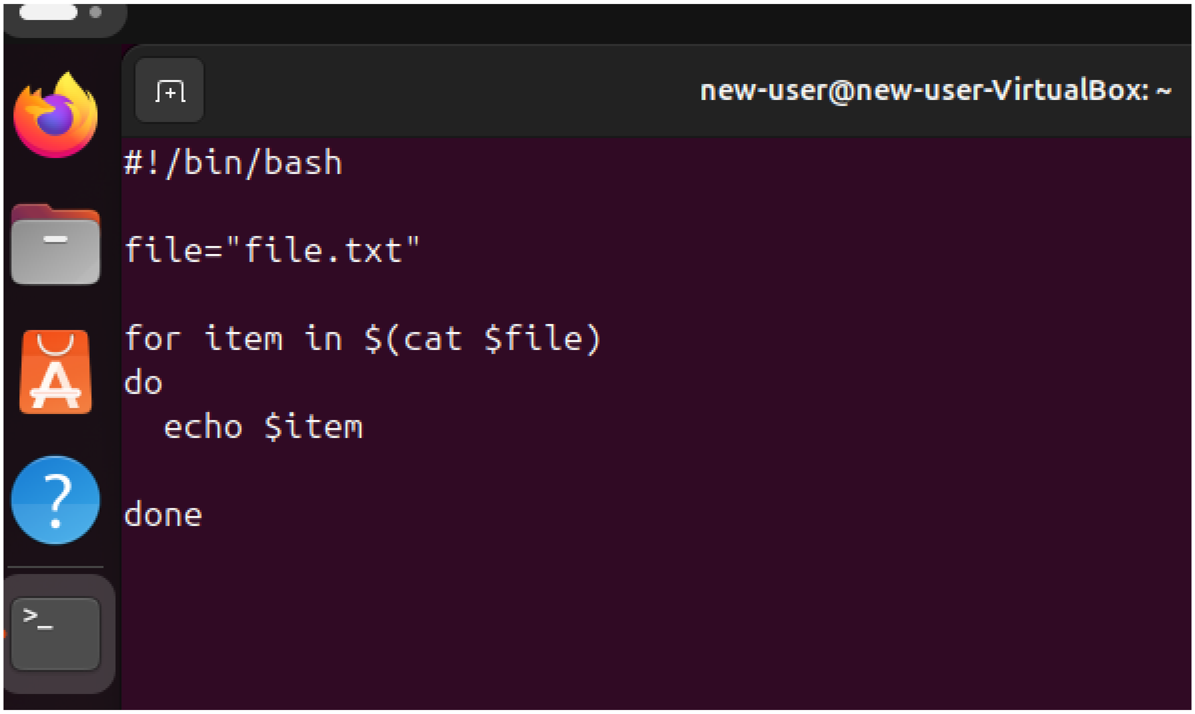
TASK#6).

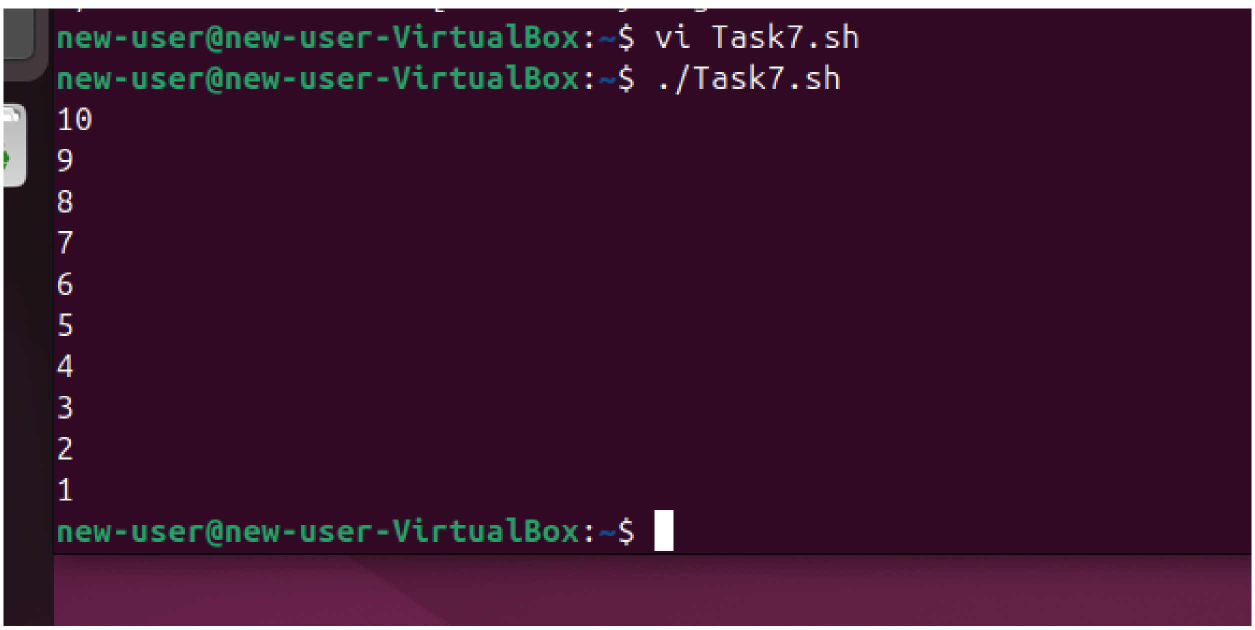





Mar 14 01:29
new-user@new-user-VirtualBox: ~
GNU nano 7.2 file.txt *
Mango
Grapes
Hello
Hello
Das
Hello


Mar 14 01:3
new-user@new-user-VirtualBox: ~
new-user@new-user-VirtualBox:~$ vi Task7.sh
new-user@new-user-VirtualBox:~$ nano file.txt
new-user@new-user-VirtualBox:~$ vi Task7.sh
[1]+ Stopped vi Task7.sh
new-user@new-user-VirtualBox:~$ vi Task7.sh
new-user@new-user-VirtualBox:~$ ./Task7.sh
Mango
Grapes
Hello
Hello
Das
Hello
new-user@new-user-VirtualBox:~$


new-user@new-user-VirtualBox: ~
#!/bin/bash
file="file.txt"
for item in $(cat $file)
do
echo $item
done

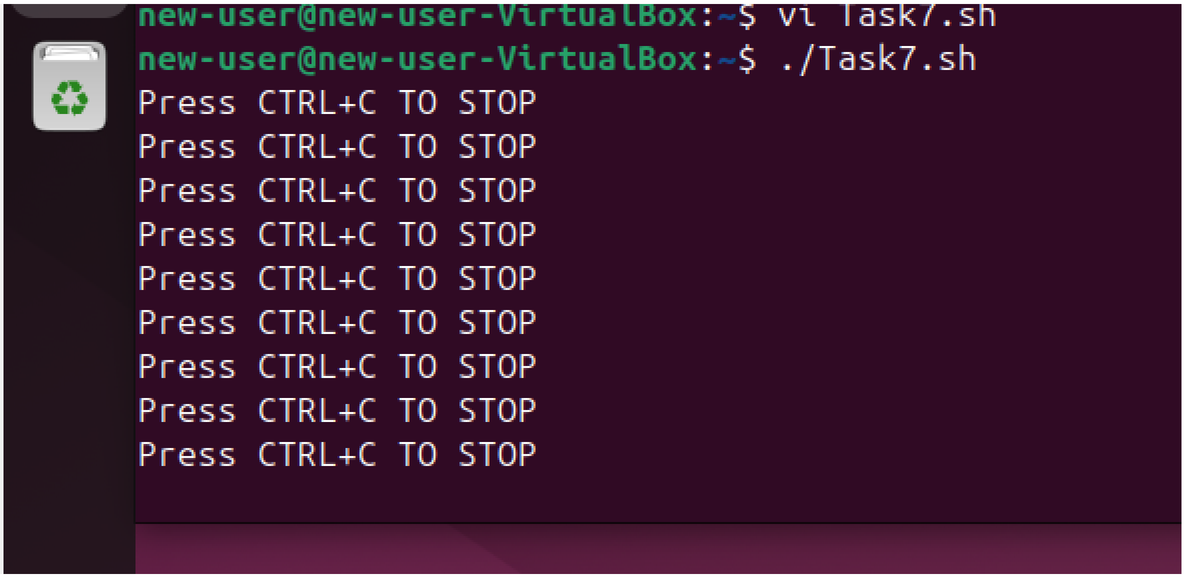

new-user@new-user-VirtualBox:~$ vi Task7.sh
new-user@new-user-VirtualBox:~$ ./Task7.sh
10
9
8
7
6
5
4
3
2
1
new-user@new-user-VirtualBox:~$

```
new-user@new-user-Virtu

#!/bin/bash



num=10


while [ $num -gt 0 ]
do
  echo $num

  let num--




done
```

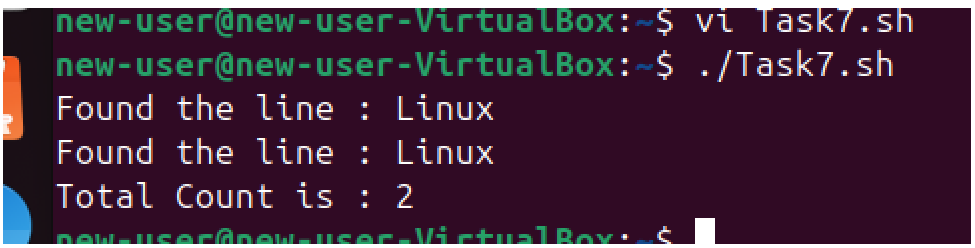
```
new-user@new-user-VirtualBox:~$ vi Task7.sh
new-user@new-user-VirtualBox:~$ ./Task7.sh
1
2
3
4
5
new-user@new-user-VirtualBox:~$
```

```
#!/bin/bash

i=1
until [ $i -gt 5 ]
do

 echo $i



 let i++


done
```

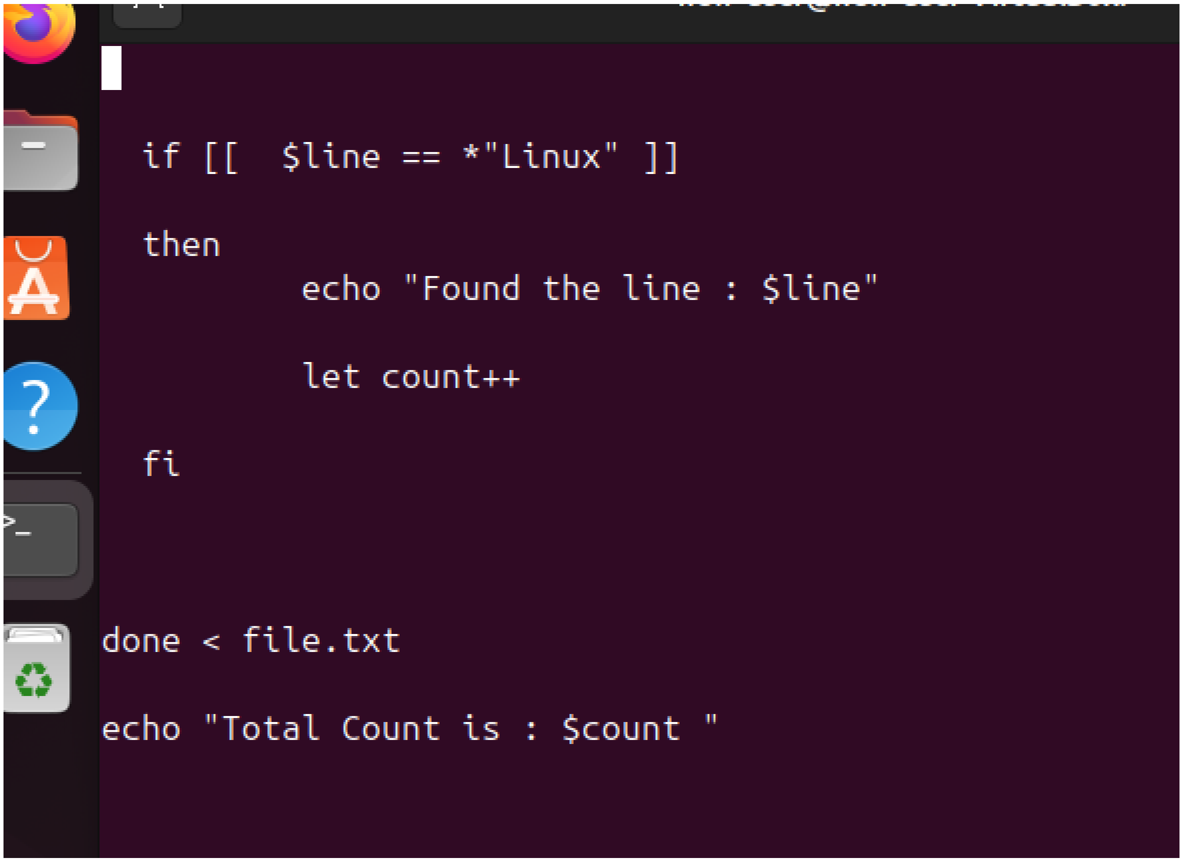
```
#!/bin/bash


while true
do
  echo "Press CTRL+C TO STOP "
  sleep 2
done
```

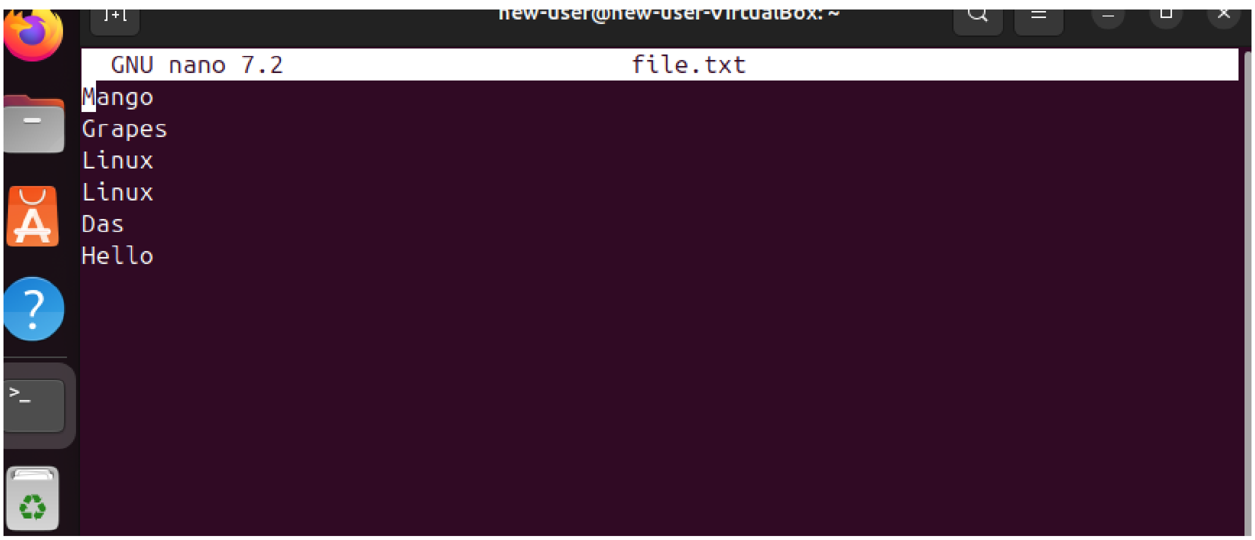

new-user@new-user-VirtualBox:~$ vi Task7.sh
new-user@new-user-VirtualBox:~$ ./Task7.sh
Press CTRL+C TO STOP
Press CTRL+C TO STOP
Press CTRL+C TO STOP
Press CTRL+C TO STOP
Press CTRL+C TO STOP
Press CTRL+C TO STOP
Press CTRL+C TO STOP
Press CTRL+C TO STOP
Press CTRL+C TO STOP

TASK#7).

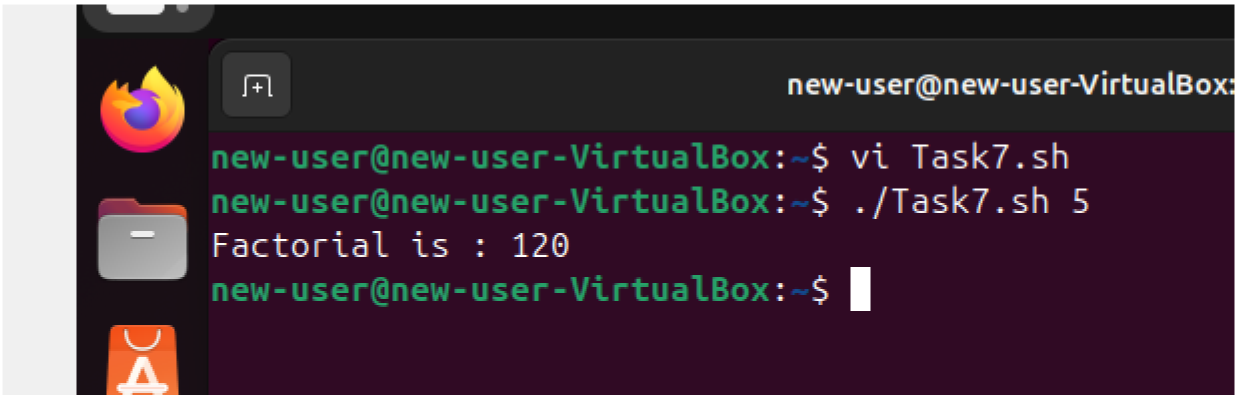

new-user@new-user-VirtualBox:~$ vi Task7.sh
new-user@new-user-VirtualBox:~$ ./Task7.sh
Found the line : Linux
Found the line : Linux
Total Count is : 2

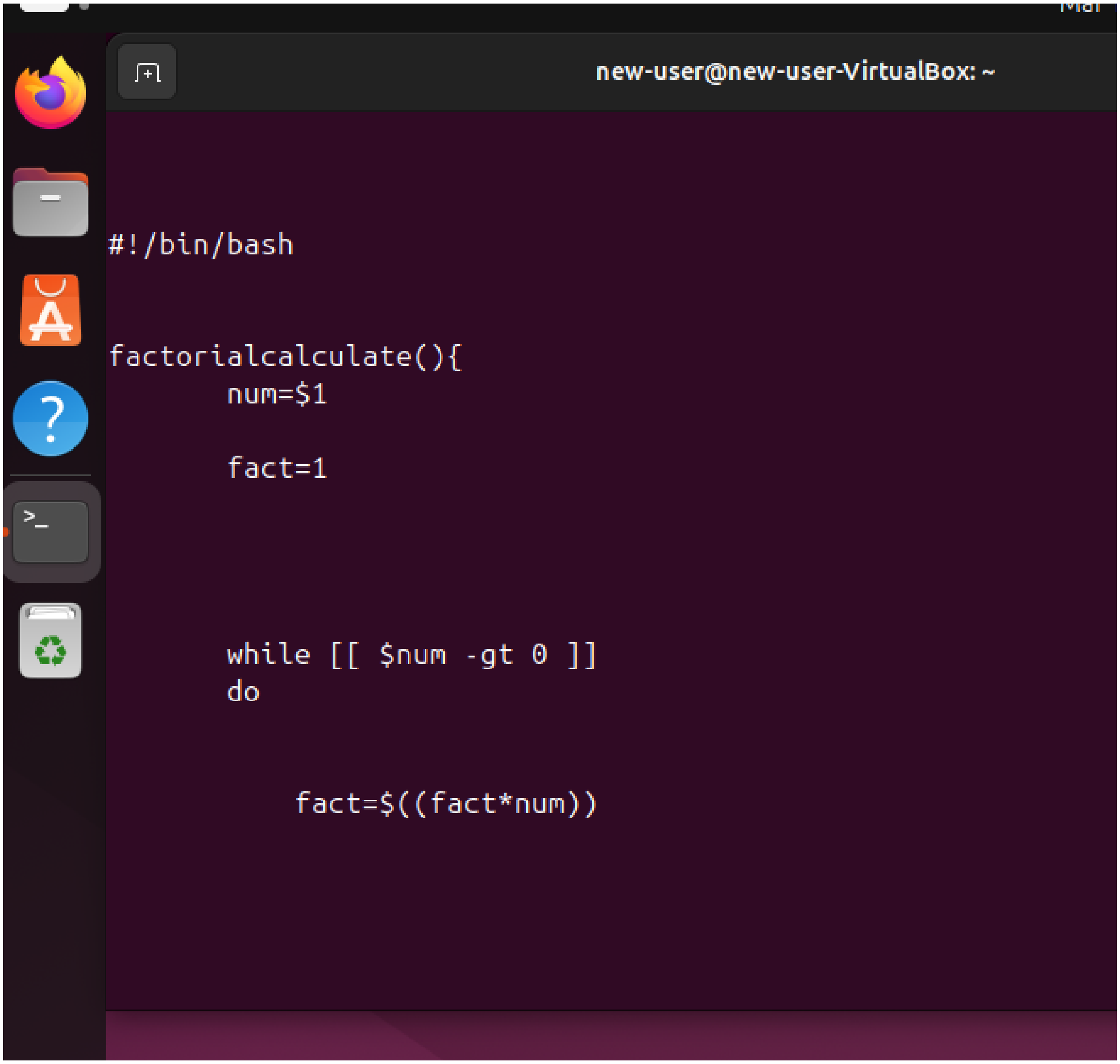
```
#!/bin/bash


file=file.txt


count=0
while read line
do








  if [[  $line == *"Linux" ]]

  then
          echo "Found the line : $line"
```


if [[$line == *"Linux"]]
then
echo "Found the line : $line"
let count++
fi
done < file.txt
echo "Total Count is : $count "


new-user@new-user-VirtualBox: ~
GNU nano 7.2
file.txt
Mango
Grapes
Linux
Linux
Das
Hello

TASK#8).


new-user@new-user-VirtualBox:
new-user@new-user-VirtualBox:~$ vi Task7.sh
new-user@new-user-VirtualBox:~$ ./Task7.sh 5
Factorial is : 120
new-user@new-user-VirtualBox:~$


new-user@new-user-VirtualBox: ~
#!/bin/bash
factorialcalculate(){
num=$1
fact=1
while [[$num -gt 0]]
do
fact=$((fact*num))

new-user@new-user-VirtualBox: ~

```
       while [[ $num -gt 0 ]]
       do


           fact=$((fact*num))




            let num--

      done



 echo "Factorial is : $fact"
}
factorialcalculate $1
```

TASK#9).

```
new-user@new-user-VirtualBox:~$ chmod +x rename_files.sh
new-user@new-user-VirtualBox:~$ ./rename_files.sh . file
Starting file renaming in directory: .
New naming pattern: fileN.ext
Renamed: a.txt -> file1.txt
Renamed: b.txt -> file2.txt
Renamed: c.txt -> file3.txt
Renamed: file1. -> file4.
```

```
Renamed: file10. -> file5.
Renamed: file11. -> file6.
Renamed: file12. -> file7.
Renamed: file13. -> file8.
Renamed: file14. -> file9.
Renamed: file15. -> file10.
Renamed: file16. -> file11.
Renamed: file17. -> file12.
Renamed: file18. -> file13.
Renamed: file19. -> file14.
Renamed: file2. -> file15.
Renamed: file20. -> file16.
Renamed: file21. -> file17.
Renamed: file22. -> file18.
Renamed: file23. -> file19.
Renamed: file24. -> file20.
Renamed: file25. -> file21.
Renamed: file26. -> file22.
Renamed: file27 -> file23
Renamed: file27. -> file24.
Renamed: file28. -> file25.
Renamed: file29. -> file26.
Renamed: file3. -> file27.
Renamed: file30. -> file28.
Renamed: file31. -> file29.
Renamed: file32. -> file30.
Renamed: file33. -> file31.
Renamed: file34. -> file32.
Renamed: file35. -> file33.
Renamed: file36. -> file34.
Renamed: file4. -> file35.
Renamed: file5. -> file36.
Renamed: file6. -> file37.
```

```
Renamed: file2. -> file15.
Renamed: file20. -> file16.
Renamed: file21. -> file17.
Renamed: file22. -> file18.
Renamed: file23. -> file19.
Renamed: file24. -> file20.
Renamed: file25. -> file21.
Renamed: file26. -> file22.
Renamed: file27 -> file23
Renamed: file27. -> file24.
Renamed: file28. -> file25.
Renamed: file29. -> file26.
Renamed: file3. -> file27.
Renamed: file30. -> file28.
Renamed: file31. -> file29.
Renamed: file32. -> file30.
Renamed: file33. -> file31.
Renamed: file34. -> file32.
Renamed: file35. -> file33.
Renamed: file36. -> file34.
Renamed: file4. -> file35.
Renamed: file5. -> file36.
Renamed: file6. -> file37.
Renamed: file7. -> file38.
Renamed: file8. -> file39.
Renamed: file9. -> file40.
Renamed: rename_files.sh -> file41.sh
Renaming complete.
Total files processed: 41
Errors encountered: 0
```

I had different files in my folder but first when I runned my code it just converted the name but the file extension was not conserved so that's why there are files like file5. Etc.

```bash
#!/bin/bash


if [ "$#" -ne 2 ]; then
    echo "Usage: $0 <directory_path> <new_file_name_pattern>"
    exit 1
fi

DIR="$1"
PATTERN="$2"


if [ ! -d "$DIR" ]; then
    echo "Error: Directory '$DIR' does not exist."
    exit 1
fi

count=1
errors=0

echo "Starting file renaming in directory: $DIR"
echo "New naming pattern: ${PATTERN}N.ext"



for file in "$DIR"/*; do

    if [ -f "$file" ]; then
        filename=$(basename "$file")
        extension="${filename##*.}"
```

new-user@new-user-VirtualBox: ~

```
for file in "$DIR"/*; do

    if [ -f "$file" ]; then
        filename=$(basename "$file")
        extension="${filename##*.}"


        if [ "$filename" = "$extension" ]; then
            newname="${PATTERN}${count}"
        else
            newname="${PATTERN}${count}.${extension}"
        fi

        newpath="$DIR/$newname"


        if mv "$file" "$newpath" 2>/dev/null; then
            echo "Renamed: $filename -> $newname"
        else
            echo "Error renaming: $filename"
            ((errors++))
        fi

        ((count++))
    fi
done


echo "Renaming complete."
```

new-user@new-u

```
    if [ -f "$file" ]; then
        filename=$(basename "$file")
        extension="${filename##*.}"


        if [ "$filename" = "$extension" ]; then
            newname="${PATTERN}${count}"
        else
            newname="${PATTERN}${count}.${extension}"
        fi

        newpath="$DIR/$newname"


        if mv "$file" "$newpath" 2>/dev/null; then
            echo "Renamed: $filename -> $newname"
        else
            echo "Error renaming: $filename"
            ((errors++))
        fi

        ((count++))
    fi
done


echo "Renaming complete."
echo "Total files processed: $((count-1))"
echo "Errors encountered: $errors"
```

TASK#10).

```bash
#!/bin/bash


if [ "$#" -ne 2 ]; then


      echo " Usage: $0 <directory_path> <days_old>"

      exit 1
fi

DIR="$1"
DAYS="$2"


if [ ! -d "$DIR" ]; then

    echo " Error: Directory '$DIR' does not exist."
    exit 1
fi


echo " Starting cleanup in: $DIR"

echo " Removing files older than $DAYS days..."



files_before=$(find "$DIR" -type f -mtime +"$DAYS" | wc -l)
```

new-user@new-user-VirtualBox: ~

```
    echo " Error: Directory '$DIR' does not exist."
    exit 1
fi


echo " Starting cleanup in: $DIR"

echo " Removing files older than $DAYS days..."



files_before=$(find "$DIR" -type f -mtime +"$DAYS" | wc -l)


find "$DIR" -type f -mtime +"$DAYS" -exec rm -f {} \;


dirs_before=$(find "$DIR" -type d -empty | wc -l)
find "$DIR" -type d -empty -delete



echo " Cleanup complete!"

echo " Files removed: $files_before"

echo " Empty directories removed: $dirs_before"
```

```
file12.     file16.  file1.txt  file23   file27.  file30.    file34.
new-user@new-user-VirtualBox:~$ ./cleanup.sh . 7
 Starting cleanup in: .
 Removing files older than 7 days...
 Cleanup complete!
 Files removed: 386
 Empty directories removed: 169
new-user@new-user-VirtualBox:~$
```